



Name: Ajayi Oluwapelumi Joseph

Matriculation Number: AUL/CMP/22/013

Course Code: CMP 412

Course Title: Computer Network and Communication

Question:

Write a code to show the communication between the client and the server side (1 server - 1 client). In your code, the client should ask for time (Add time).

Server.py

```
import socket
import datetime # Imported to handle time functions

ss = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)
print("Server started. Waiting for connection...")

ip = socket.gethostname()
port = 9000

ss.bind((ip, port))
ss.listen(1) # Listening for 1 client

con, addr = ss.accept()
print("Client connected from: ", addr)

while True:
    # Receive message from client
    data = con.recv(1024).decode()

    if not data or data.lower() == "bye":
        print("Client ended the conversation.")
        break
```

```

print(f"Client asked: {data}")

# Logic to check if client is asking for time
if "time" in data.lower():
    # Get current time
    current_time = datetime.datetime.now().strftime("%H:%M:%S")
    reply = f"The current time is: {current_time}"
else:
    reply = "Please ask me for the 'time'."

# Send the reply back to the client
con.sendall(bytes(reply.encode("ascii")))
print("Response sent to client.")

con.close()
ss.close()

```

```

import socket

cs = socket.socket(family=socket.AF_INET, type=socket.SOCK_STREAM)

ip = socket.gethostname()
port = 9000

try:
    cs.connect((ip, port))
    print("Connected to server.")
except:
    print("Could not connect. Make sure the server is running first.")
    exit()

msg = ""

while msg.lower() != "bye":
    # Ask user for input matching the assignment requirements
    msg = input("\nAsk the server for 'time' (or type 'bye' to quit): ")

    # Send message to server
    cs.sendall(bytes(msg.encode("ascii")))

    if msg.lower() == "bye":
        break

    # Receive message from server
    data = cs.recv(1024).decode()

    print("Server replied: ", data)

```

```
cs.close()
```

Output:

```
PS C:\Users\Pelumi of God\Documents\400-Level-First-Semester\CMP 412\assignment> py .\client_assignment.py
Connected to server.

Ask the server for 'time' (or type 'bye' to quit): time
Server replied: The current time is: 01:47:52

Ask the server for 'time' (or type 'bye' to quit): time
Server replied: The current time is: 01:47:57

Ask the server for 'time' (or type 'bye' to quit): bye
PS C:\Users\Pelumi of God\Documents\400-Level-First-Semester\CMP 412\assignment>

PS C:\Users\Pelumi of God\Documents\400-Level-First-Semester> py .\server_assignment.py
Server started. Waiting for connection...
Client connected from: ('10.5.2.163', 5585)
Client asked: time
Response sent to client.
Client asked: time
Response sent to client.
Client ended the conversation.
PS C:\Users\Pelumi of God\Documents\400-Level-First-Semester\CMP 412\assignment>
```